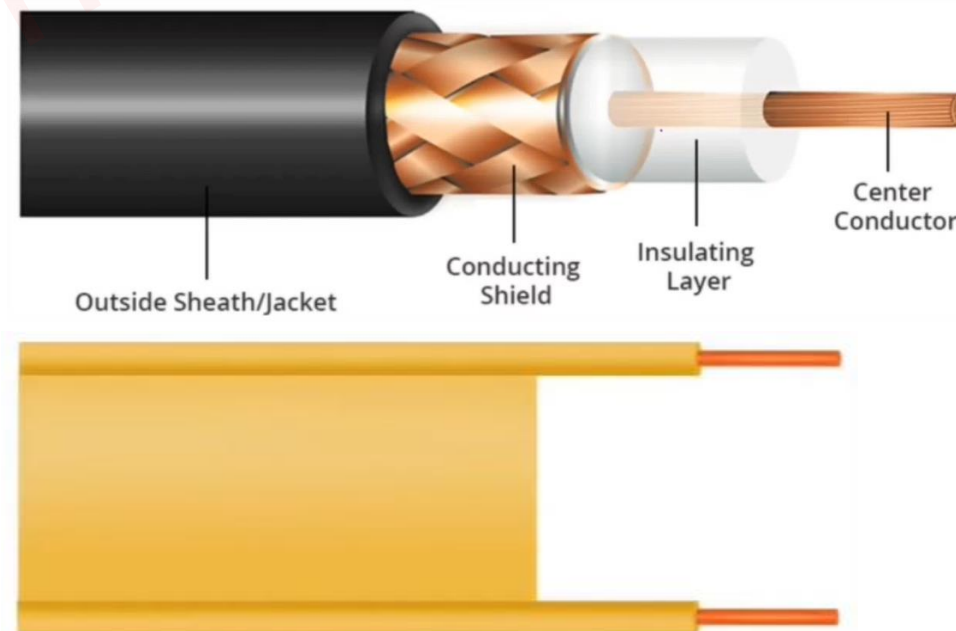


Scattering Parameters

Basics of Scattering Parameters

❑ Scattering Parameters are used for the analysis of the following Devices:

- Microwave Devices [Isolator, Directional Coupler, Circulator, etc.]
- RF Circuits [Amplifier, Filter, Mixer, Oscillator, Switch etc.]
- Antennas [Microstrip Antenna, Dipole Antenna, Horn Antenna etc.]
- Waveguides [Circular Waveguide and Rectangular Waveguide]
- Transmission Lines [Two Parallel wires, Microstrip, Coaxial, etc.]



Why to use Scattering Parameters at Microwave Frequencies?

- ❑ At Microwave Frequencies, It is not possible to provide proper Open Circuits and Short circuits to the port of Microwave devices.
- ❑ Equipment is not available to measure instantaneous Voltage and Current at Microwave Frequencies.
- ❑ Active Devices have stability issues at Microwave frequencies. [Microwave Transistor, Microwave Diodes etc.]
- ❑ Microwave Devices have frequency-dependent characteristics.
- ❑ For the measurement of H, Y, Z, and ABCD parameters, lower frequencies are suitable.

How to measure Scattering Parameters?

□ In measurement of Scattering Parameters S_{ij} , i is the output port and j is the input port.

$$S_{ij} = \frac{\text{Normlized output wave at } i^{\text{th}} \text{ port}}{\text{Normlized input wave at } j^{\text{th}} \text{ port}} = \frac{b_i}{a_j}$$

Where,

$$b_i = \frac{\text{Output Voltage at } i^{\text{th}} \text{ port}}{\sqrt{Z_{0i}}}$$

$$a_j = \frac{\text{Input Voltage at } j^{\text{th}} \text{ port}}{\sqrt{Z_{0j}}}$$



- S_{11} is the Reflection Coefficient at port 1.
- S_{22} is the Reflection Coefficient at port 2.
- S_{12} is the Transmission Coefficient From port 2 to port 1.
- S_{21} is the Transmission Coefficient From port 1 to port 2.

Properties of S Parameters

1. S Matrix is the Square Matrix of order $n \times n$.
 - Where n is a number of ports with a given device.

⇒ E.g. - For three port device

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

2. Identity Property.

$$[S][S]^{\dagger} = [I]$$

⇒ For three port device

$$\Rightarrow \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \begin{bmatrix} S_{11}^{\dagger} & S_{12}^{\dagger} & S_{13}^{\dagger} \\ S_{21}^{\dagger} & S_{22}^{\dagger} & S_{23}^{\dagger} \\ S_{31}^{\dagger} & S_{32}^{\dagger} & S_{33}^{\dagger} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. For symmetrical Devices.

- This property applies to symmetrical ports only.

$$S_{ij} = S_{ji}$$

4. A Network is Reciprocal If,

$$[S] = [S]^T$$

⇒ For three port n/w

$$\Rightarrow \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} = [S]$$

$$\Rightarrow [S]^T = \begin{bmatrix} S_{11} & S_{21} & S_{31} \\ S_{12} & S_{22} & S_{32} \\ S_{13} & S_{23} & S_{33} \end{bmatrix}$$

5. A Network is Lossless If,

$$\Rightarrow [S]^T [S]^\dagger = [I]$$

~> For two port N/w.

$$\Rightarrow [S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \Rightarrow [S]^T = \begin{bmatrix} S_{11} & S_{21} \\ S_{12} & S_{22} \end{bmatrix}, \Rightarrow [S]^\dagger = \begin{bmatrix} S_{11}^\dagger & S_{12}^\dagger \\ S_{21}^\dagger & S_{22}^\dagger \end{bmatrix}$$

~> Manual

$$\Rightarrow \begin{bmatrix} S_{11} & S_{21} \\ S_{12} & S_{22} \end{bmatrix} \begin{bmatrix} S_{11}^\dagger & S_{12}^\dagger \\ S_{21}^\dagger & S_{22}^\dagger \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sim |S_{11}|^2 + |S_{21}|^2 = 1$$

$$S_{11} S_{12}^\dagger + S_{21} S_{22}^\dagger = 0$$

$$S_{12} S_{11}^\dagger + S_{22} S_{21}^\dagger = 0$$

$$|S_{12}|^2 + |S_{22}|^2 = 1$$

~> In general

$$\Rightarrow |S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 + \dots + |S_{1n}|^2 = 1$$

[rowwise]

$$\Rightarrow |S_{11}|^2 + |S_{21}|^2 + |S_{31}|^2 + \dots + |S_{n1}|^2 = 1$$

[columnwise]

6. Multiplication of Two Rows or Columns

$$\Rightarrow \sum_{i=1}^n s_{ij} s_{ik}^T = 0$$

$$j \neq k$$

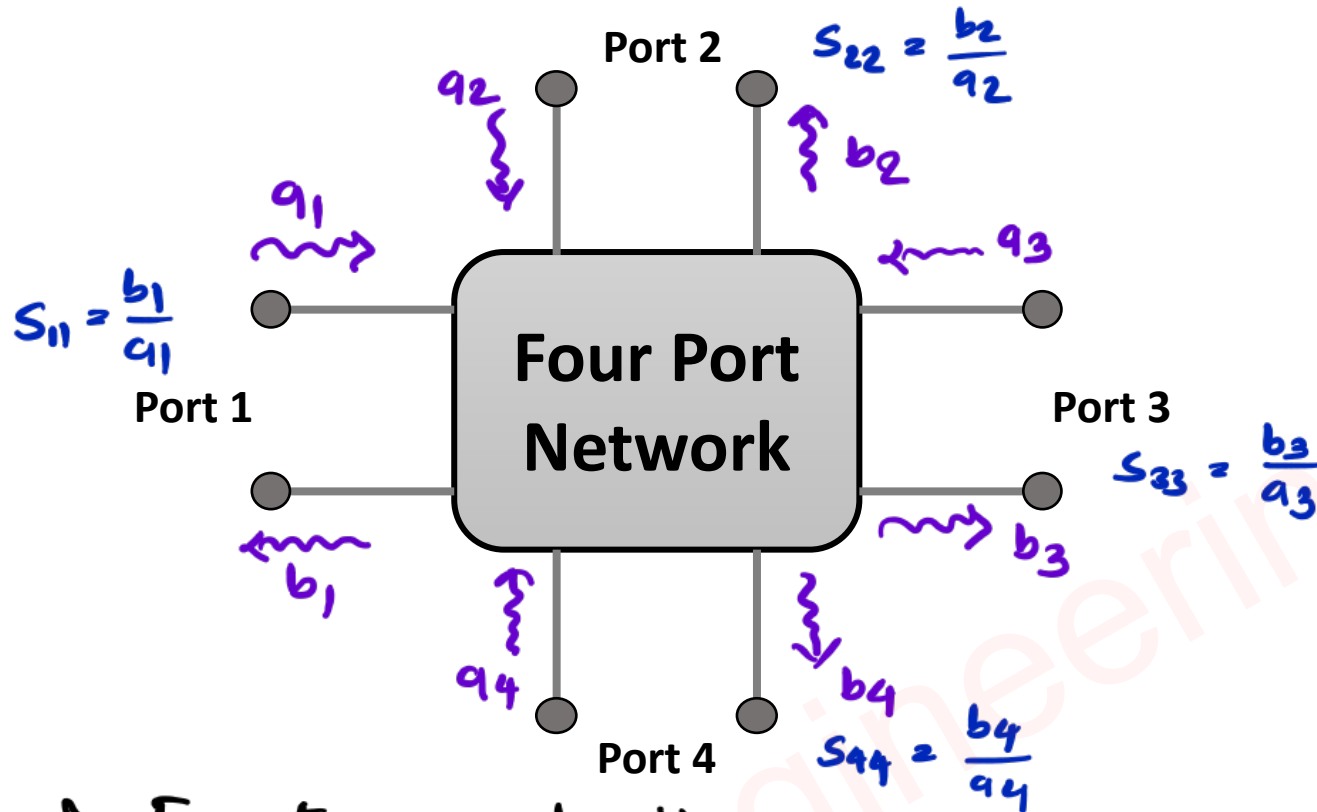
[Row multiplication]

$$\Rightarrow \sum_{j=1}^n s_{ij} s_{kj}^T = 0$$

$$i \neq k$$

[Columns multiplication]

S Parameters of Four Port N/W



Magic Tee
{4 Port N/W}

∴ For four port N/W

$$\Rightarrow [S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix}$$

~> Basic form of S_{ij}

$$\Rightarrow S_{ij} = \frac{\text{Normalized Output wave at } i^{\text{th}} \text{ port}}{\text{Normalized Input wave at } j^{\text{th}} \text{ port}} = \frac{b_i}{a_j}$$

$$\Rightarrow S_{11} = \frac{b_1}{a_1}, \quad S_{12} = \frac{b_1}{a_2}, \quad S_{13} = \frac{b_1}{a_3}, \quad S_{14} = \frac{b_1}{a_4}$$

~> So, total output at port 1 (b_1)

$$\rightarrow b_1 = a_1 S_{11} + a_2 S_{12} + a_3 S_{13} + a_4 S_{14} \quad \text{--- (1)}$$

~> Similarly,

$$\rightarrow b_2 = a_1 S_{21} + a_2 S_{22} + a_3 S_{23} + a_4 S_{24} \quad \text{--- (2)}$$

$$\rightarrow b_3 = a_1 S_{31} + a_2 S_{32} + a_3 S_{33} + a_4 S_{34} \quad \text{--- (3)}$$

$$\rightarrow b_4 = a_1 S_{41} + a_2 S_{42} + a_3 S_{43} + a_4 S_{44} \quad \text{--- (4)}$$

~> Using, eq.ⁿ (1), (2), (3) and (4)

$$\Rightarrow [b] = [S][a]$$

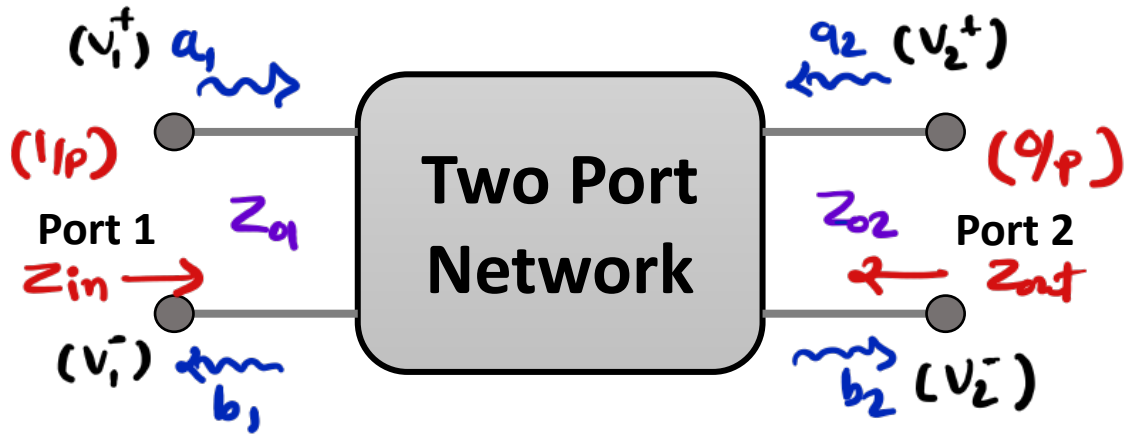
$$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

Return loss

Return loss parameters at port 1, 2, 3 & 4

$$S_{11} = \frac{b_1}{a_1}, \quad S_{22} = \frac{b_2}{a_2}, \quad S_{33} = \frac{b_3}{a_3}, \quad S_{44} = \frac{b_4}{a_4}$$

S Parameters of Two Port N/W



$$\Rightarrow a_1 = \frac{V_1^+}{\sqrt{Z_{o1}}}, \quad b_1 = \frac{V_1^-}{\sqrt{Z_{o1}}}, \quad a_2 = \frac{V_2^+}{\sqrt{Z_{o2}}}, \quad b_2 = \frac{V_2^-}{\sqrt{Z_{o2}}}$$

→ Scattering matrix

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

→ Basic form S_{ij}

$$S_{ij} = \frac{\text{Normalized Output wave at } i^{\text{th}} \text{ port}}{\text{Normalized Input wave at } j^{\text{th}} \text{ port}} = \frac{b_i}{a_j}$$

$$\Rightarrow S_{11} = \frac{b_1}{a_1}, \quad S_{12} = \frac{b_1}{a_2}, \quad S_{21} = \frac{b_2}{a_1}, \quad S_{22} = \frac{b_2}{a_2}$$

$\Rightarrow$ Total Output wave at port 1 and port 2

$$\Rightarrow b_1 = a_1 S_{11} + a_2 S_{12}$$

$$\Rightarrow b_2 = a_1 S_{21} + a_2 S_{22}$$

$\Rightarrow$ In form of matrix

$$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$\Rightarrow$ Scattering parameters in form of voltage.

$$\Rightarrow S_{11} = \frac{b_1}{a_1} = \frac{V_1^- / \sqrt{Z_{01}}}{V_1^+ / \sqrt{Z_{01}}} = \frac{V_1^-}{V_1^+}$$

$$\Rightarrow S_{12} = \frac{b_1}{a_2} = \frac{V_1^- / \sqrt{Z_{01}}}{V_2^+ / \sqrt{Z_{02}}} = \sqrt{\frac{Z_{02}}{Z_{01}}} \left(\frac{V_1^-}{V_2^+} \right)$$

$$\Rightarrow S_{21} = \frac{b_2}{a_1} = \frac{V_2^- / \sqrt{Z_{02}}}{V_1^+ / \sqrt{Z_{01}}} = \sqrt{\frac{Z_{01}}{Z_{02}}} \left(\frac{V_2^-}{V_1^+} \right)$$

$$\Rightarrow S_{22} = \frac{b_2}{a_2} = \frac{V_2^- / \sqrt{Z_{02}}}{V_2^+ / \sqrt{Z_{02}}} = \frac{V_2^-}{V_2^+}$$

~> If $Z_{01} = Z_{02}$

$$\sim S_{11} = \frac{V_1^-}{V_1^+}, \quad S_{12} = \frac{V_1^-}{V_2^+}, \quad S_{21} = \frac{V_2^-}{V_1^+}, \quad S_{22} = \frac{V_2^-}{V_2^+}$$

~> Total Output voltage V_1^- and V_2^-

$$\Rightarrow V_1^- = V_1^+ S_{11} + V_2^+ S_{12}$$

$$\Rightarrow V_2^- = V_1^+ S_{21} + V_2^+ S_{22}$$

→ At Port 1 Reflection Coefficient

$$\Gamma_1 = \frac{Z_{in} - Z_{01}}{Z_{in} + Z_{01}} = \frac{b_1}{a_1} = \frac{V_1^- / \sqrt{Z_{01}}}{V_1^+ / \sqrt{Z_{01}}} = \frac{V_1^-}{V_1^+}$$

→ At Port 2 Reflection Coefficient

$$\Gamma_2 = \frac{Z_{out} - Z_{02}}{Z_{out} + Z_{02}} = \frac{b_2}{a_2} = \frac{V_2^- / \sqrt{Z_{02}}}{V_2^+ / \sqrt{Z_{02}}} = \frac{V_2^-}{V_2^+}$$

Examples of S Parameters

Question 1 – The S Matrix of a two-port network is given by:

$$[S] = \begin{bmatrix} 0.2 \angle 0 & 0.9 \angle 90 \\ 0.9 \angle 90 & 0.1 \angle 90 \end{bmatrix}$$

Find the given network is lossless and reciprocal or not.

$$\Rightarrow [S]^T = \begin{bmatrix} 0.2 \angle 0 & 0.9 \angle 90 \\ 0.9 \angle 90 & 0.1 \angle 90 \end{bmatrix}$$

$\Rightarrow [S] = [S]^T$, Hence given N/w is Reciprocal N/w.

$\Rightarrow$ For lossless N/w, Cond.ⁿ is

$$\Rightarrow |S_{11}|^2 + |S_{12}|^2 = 1$$

$$\rightarrow 0.2^2 + 0.9^2 = 0.85 \neq 1$$

$\Rightarrow$ Hence, given N/w is lossy N/w.

Question 2 – The S Matrix of a Four-port network is given by:

$$[S] = \begin{bmatrix} 0.1\angle 90 & 0.8\angle 45 & 0.3\angle 90 & 0 \\ 0.8\angle 45 & 0 & 0 & 0.4\angle 45 \\ 0.3\angle 45 & 0 & 0 & 0.6\angle 45 \\ 0 & 0.4\angle 45 & 0.6\angle 45 & 0 \end{bmatrix}$$

Find the given network is lossless and reciprocal or not.

$$\Rightarrow [S]^T = \begin{bmatrix} 0.1\angle 90 & 0.8\angle 45 & \underline{0.3\angle 45} & 0 \\ 0.8\angle 45 & 0 & 0 & 0.4\angle 45 \\ \underline{0.3\angle 90} & 0 & 0 & 0.6\angle 45 \\ 0 & 0.4\angle 45 & 0.6\angle 45 & 0 \end{bmatrix}$$

$$\Rightarrow [S] \neq [S]^T$$

$\Rightarrow$ Hence, given N/w is not reciprocal N/w.

$\Rightarrow$ For lossless N/w, Cond.ⁿ is

$$\Rightarrow |S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 + |S_{14}|^2 = 1$$

$$\Rightarrow 0.1^2 + 0.8^2 + 0.3^2 + 0^2 = 0.74 \neq 1$$

$\Rightarrow$ Hence, given N/w is lossy N/w.

Examples of S Parameters

Question 3 – The S Matrix of a two-port network is given by:

$$[S] = \begin{bmatrix} 0.15 \angle 0 & 0.85 \angle -45 \\ 0.85 \angle 45 & 0.2 \angle 0 \end{bmatrix}$$

Find,

1. Network is reciprocal?
2. Network is Lossless?
3. If port 2 is terminated with a matched load, find return loss at port 1.
4. If port 2 is terminated with a Short Circuit, find return loss at port 1.

$$\Rightarrow [S]^T = \begin{bmatrix} 0.15 \angle 0 & 0.85 \angle 45 \\ 0.85 \angle -45 & 0.2 \angle 0 \end{bmatrix}$$

$$\Rightarrow [S] \neq [S]^T$$

$\Rightarrow$ So, given N/w is not reciprocal N/w.

~> For lossless N/w, Cond.ⁿ is

$$\Rightarrow |S_{11}|^2 + |S_{12}|^2 = 1$$

$$\rightarrow 0.15^2 + 0.85^2 = 0.745 \neq 1$$

~> Hence, given N/w is lossy N/w.

~> Reflection coefficient at port 1

$$\Gamma_1 = \frac{b_1}{a_1}$$

$$\sim \text{Return Loss}_1 = 20 \log |\Gamma_1|$$

~> For two port N/w

$$b_1 = a_1 S_{11} + a_2 S_{12}$$

$$b_2 = a_1 S_{21} + a_2 S_{22}$$

~> If port 2 is terminated by matched load,

$$\boxed{a_2 = 0}$$

$$\Rightarrow \frac{b_1}{a_1} = S_{11} = \Gamma_1 = 0.15$$

$$\Rightarrow R.L._2 = 20 \log |r_1| = 20 \log |0.15| = -16.48 \text{ dB}$$

$\sim$ If port 2 is terminated by short circuit,

$$\boxed{a_2 = -b_2}$$

$$\Rightarrow b_1 = a_1 S_{11} - b_2 S_{12} \quad \text{--- (2)}$$

$$\Rightarrow b_2 = a_1 S_{21} - b_2 S_{22} \Rightarrow b_2 (1 + S_{22}) = a_1 S_{21}$$

$$\Rightarrow b_2 = a_1 \left[\frac{S_{21}}{1 + S_{22}} \right] \quad \text{--- (1)}$$

$\sim$ Put eq. (1) into (2)

$$\Rightarrow b_1 = a_1 S_{11} - a_1 \left[\frac{S_{12} S_{21}}{1 + S_{22}} \right]$$

$$\Rightarrow \frac{b_1}{a_1} = S_{11} - \frac{S_{12} S_{21}}{1 + S_{22}} = r_1$$

$$\Rightarrow r_1 = 0.15 - \frac{0.85 \angle -45^\circ \times 0.85 \angle 45^\circ}{1 + 0.2}$$

$$\Rightarrow r_1 = 0.15 - \frac{0.85^2}{1.2}$$

$$\Rightarrow r_1 = -0.45$$

$$\begin{aligned}\rightarrow R.L._2 &= 20 \log |r_1| \\ &= 20 \log 0.45 \\ &= -6.93 \text{ dB}\end{aligned}$$

Engineering Funda